

# JEE MAIN REVISION MASTER: QUADRATIC EQUATIONS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                               | Frequency | Category          |
|---------------|--------------------------------------------|-----------|-------------------|
| Standard      | Location of Roots & Quadratic Inequalities | 2 PYQs    | Location of Roots |
| Standard      | Roots of Quadratic & Vieta's Relations     | 1 PYQs    | Quadratic Roots   |

## 2. Master Formula Reference Card

| Concept / Topic                            | Mathematical Formulation & Governing Equations                                                                                                                         |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location of Roots & Quadratic Inequalities | $\text{Both roots} > k \iff D \geq 0, \quad -\frac{b}{2a} > k, \quad a \cdot f(k) > 0$                                                                                 |
| Roots of Quadratic & Vieta's Relations     | $\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad  \alpha - \beta  = \frac{\sqrt{D}}{ a }, \quad aS_n + bS_{n-1} + cS_{n-2} = 0 \text{ (Newton)}$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Location of Roots & Quadratic Inequalities (Standard Yield — 2 PYQs)

**Core Principle:** Conditions for roots relative to real numbers  $k$ ,  $k_1$ ,  $k_2$ , and signs of quadratic expression  $ax^2 + bx + c$ .

■ **Caution / JEE Pitfall:** For both roots to be greater than  $k$ , checking  $a \cdot f(k) > 0$  alone is NOT sufficient; you must ALSO check  $D \geq 0$  and  $-\frac{b}{2a} > k$ .

■ **Speed Trick:** If  $a > 0$ , the quadratic  $f(x) = ax^2 + bx + c > 0$  for all real  $x$  if and only if  $D < 0$ .

**Representative Source PYQs:**

- [2023 Shift] Q#2 (MCQ): "Given that  $a, b$  and  $x$  are real numbers and  $a < b, x < 0$ , then..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "If  $\alpha$  and  $\beta$  are roots of the equation  $x^2 = x + 1 = 0$ , then  $\alpha^2 + \beta^2$  is..." [Ans: N/A]

### 3.2 Roots of Quadratic & Vieta's Relations (Standard Yield — 1 PYQs)

**Core Principle:** Discriminant analysis for real/complex roots, symmetric root functions, and Newton's power sum theorem.

■ **Caution / JEE Pitfall:** If coefficients  $a, b, c$  are rational and  $D$  is not a perfect square, roots occur in irrational conjugate pairs ( $p \pm \sqrt{q}$ ).

■ **Speed Trick:** Newton's theorem: If  $\alpha, \beta$  are roots of  $ax^2 + bx + c = 0$  and  $S_n = \alpha^n + \beta^n$ , then  $a S_n + b S_{n-1} + c S_{n-2} = 0$ .

**Representative Source PYQs:**

• **[2024 Shift] Q#1 (MCQ):** "If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2024 (Q#1)

If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is

- (A)  $x^2 - 10x - 16 = 0$
- (B)  $x^2 + 10x + 16 = 0$
- (C)  $x^2 + 10x - 16 = 0$
- (D)  $x^2 - 10x + 16 = 0$

**Correct Answer: None**

**Solution Analysis:** Let  $a$  and  $b$  be the roots of the quadratic equation. Then, the quadratic equation is  $x^2 - (a+b)x + ab = 0$ .... (i)  
It is given that  $AM=5$  and  $GM=4$  i.e.  $(a+b)/2=5$  and  $\sqrt{ab}=4 \Rightarrow a+b=10$  and  $ab=16$ . On substituting these values in Eq. (i), we obtain. Hence,  $x^2 - 10x + 16 = 0$  is the required equation.

### Benchmark Problem 2: KCET 2023 (Q#2)

Given that  $a, b$  and  $x$  are real numbers and  $a < b, x < 0$ , then

- (A)  $(a/x) \geq (b/x)$
- (B)  $(a/x) < (b/x)$
- (C)  $(a/x) \leq (b/x)$
- (D)  $(a/x) > (b/x)$

**Correct Answer: None**

**Solution Analysis:** Given,  $a < b$ . Dividing the above inequality by " $x$ "  $\Rightarrow (a/x) > (b/x)$  [■  $x < 0$ ] [Sign of inequality changes when its both sides are divided by a negative number]